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Batch: A-2
Division A

Subject: Programming with Java

ASSIGNMENT No : 12

TITLE : Write a Java program to create and use user-defined packages.

Problem Statement

Write a Java program to create and use user-defined packages.

Learning Objectives

To create and use user-defined packages for organizing Java programs.

Theory

Packages are used to organize related classes and interfaces into namespaces. User-defined packages improve code organization, reduce naming conflicts, and enhance reusability. They also support access protection and modular application development. Java packages simplify project management by grouping logically related components together. Large-scale software systems extensively use packages.

Output Screenshots

```
dslord@Fedora:~/Desktop/Coding Projects/Java/Java Lab$ java Exp12/Main.java
This is a example of a package!
dslord@Fedora:~/Desktop/Coding Projects/Java/Java Lab$
```

Exercise

Q1. Create a Java program by developing user-defined packages for a College Management System. Create separate packages for student and faculty details, and display their information using classes from these packages.

```
dslord@Fedora:~/Desktop/Coding Projects/Java/Java Lab$ java Exp12/Exercise/Ex1/Ex1.java
----- Student Details -----
Student ID : 101
Name       : Devansh
Course     : Computer Science

----- Faculty Details -----
Faculty ID : 201
Name       : Dr. Sharma
Department : CSE
dslord@Fedora:~/Desktop/Coding Projects/Java/Java Lab$
```

Q2. Create a Java program by developing a user-defined package named library that contains a class to store and display book details such as book ID, title, author, and price. Import the package into the main program and display the book information.

```
dslord@Fedora:~/Desktop/Coding Projects/Java/Java Lab$ java Exp12/Exercise/Ex2/Ex2.java
----- Book Details -----
Book ID : 101
Title   : Java Programming
Author  : Herbert Schildt
Price   : ₹650.0
dslord@Fedora:~/Desktop/Coding Projects/Java/Java Lab$
```

Conclusion

This assignment helped in understanding the concept and use of user-defined packages in Java. It demonstrated how packages organize related classes and interfaces into separate namespaces, improving code organization, reusability, and maintainability. The implementation enhanced my understanding of modular programming, access control, and package importing, making Java applications more structured and easier to manage. These concepts are widely used in developing desktop applications, web applications, Android applications, enterprise software, and other large-scale Java projects where efficient code organization and modular design are essential.